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Leveraging Large Language Models in Tourism: A Comparative Study of the Latest GPT Omni Models and BERT NLP for Customer Review Classification and Sentiment Analysis

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Abstract: In today's rapidly evolving digital landscape, customer reviews play a crucial role in shaping the reputation and success of hotels. Accurately analyzing and classifying the sentiment of these reviews offers valuable insights into customer satisfaction, enabling businesses to gain a competitive edge. This study undertakes a comparative analysis of traditional natural language processing (NLP) models, such as BERT and advanced large language models (LLMs), specifically GPT-4 omni and GPT-4o mini, both pre- and post-fine-tuning with few-shot learning. By leveraging an extensive dataset of hotel reviews, we evaluate the effectiveness of these models in predicting star ratings based on review content. The findings demonstrate that the GPT-4 omni family significantly outperforms the BERT model, achieving an accuracy of 67%, compared to BERT's 60.6%. GPT-4o, in particular, excelled in accuracy and contextual understanding, showcasing the superiority of advanced LLMs over traditional NLP methods. This research underscores the potential of using sophisticated review evaluation systems in the hospitality industry and positions GPT-4o as a transformative tool for sentiment analysis. It marks a new era in automating and interpreting customer feedback with unprecedented precision.

Keywords: sentiment analysis; hotel reviews; customer feedback; user-generated content; customer review classification; gpt-4 omni; few-shot learning; decision-making systems; forecasting models; tourism recommendation systems

Citation: Roumeliotis, K.I.; Tselikas, N.D.; Nasiopoulos, D.K. Leveraging Large Language Models in Tourism: A Comparative Study of the Latest GPT Omni Models and BERT NLP for Customer Review Classification and Sentiment Analysis. *Information* **2024**, *15*, 792. https://doi.org/10.3390/info15120792

Academic Editor: Rodolfo Delmonte

Received: 14 November 2024
Revised: 3 December 2024
Accepted: 5 December 2024
Published: 10 December 2024

1. Introduction

In the digital age, customer reviews have become a cornerstone in the success and reputation of businesses, particularly within the tourism and hospitality sectors. With the travel and tourism industry contributing 9.9 trillion U.S. dollars to the global GDP in 2023, and projections expecting this figure to reach 11.1 trillion U.S. dollars in 2024, exceeding pre-pandemic levels, the economic significance of tourism is undeniable [1]. Travelers increasingly rely on online reviews to make informed decisions about hotels, creating an environment where customer feedback directly influences business performance [2]. As the volume and complexity of reviews grow, artificial intelligence (AI) has emerged as a pivotal tool, offering innovative solutions for understanding and responding to this wealth of user-generated content.

Numerous studies have explored the application of machine learning (ML) techniques, such as support vector machines (SVMs) [3,4], random forests (RFs) [5], and logistic regression [6], for sentiment classification of user-generated content, demonstrating varying degrees of accuracy and effectiveness. Furthermore, traditional deep learning methods, including convolutional neural networks (CNNs) [7] and recurrent neural networks

(RNNs) [8], have shown promise in analyzing textual data by capturing contextual and sequential information. These approaches, while effective, often require significant feature engineering and manual intervention compared to the end-to-end capabilities of large language models (LLMs). Moreover, hybrid models that combine ML and DL techniques have also been investigated, offering unique advantages in handling large-scale datasets and multilingual reviews [9]. By comparing these methodologies to the transformative capabilities of LLMs, this study underscores how advancements in AI are reshaping sentiment analysis in tourism, providing a robust foundation for enhanced customer engagement and strategic decision-making.

AI, particularly with the advent of LLMs, is transforming how businesses in the tourism industry engage with customer feedback [10]. Beyond simple sentiment analysis, LLMs such as GPT-4 omni can discern nuanced emotional tones, automatically respond to reviews, and extract actionable insights that can significantly enhance customer satisfaction and profitability. These advanced models efficiently classify reviews, predict star ratings, and even provide personalized responses that resonate with customers, offering a level of automation and accuracy that traditional methods cannot match [11].

In this study, we explore the power of LLMs in sentiment analysis by comparing the performance of the BERT model with that of the GPT-4 omni family [12]. Through this comparative analysis, we emphasize the growing importance of AI-driven tools in tourism, illustrating how these technologies can elevate customer experience, foster stronger brand loyalty, and provide strategic insights for business growth. By harnessing the full potential of LLMs, the tourism industry is entering a new era of precision, where AI interprets reviews and actively contributes to the customer journey and overall business success.

The primary aim of this research is threefold: first, to assess the effectiveness and applicability of advanced AI models, particularly LLMs, within the tourism sector; second, to compare the performance of these models both before and after fine-tuning with few-shot learning; and third, to address specific research questions that previous studies have not adequately explored:

- Q1: Do NLP models or LLMs demonstrate superior efficacy in assessing user-generated content, such as hotel customer reviews, in terms of sentiment analysis and star rating prediction?
- Q2: Which model within the GPT-4 omni family exhibits superior performance before fine-tuning?
- Q3: Which model within the GPT-4 omni family demonstrates superior performance after fine-tuning with few-shot learning?
- Q4: How significant is the cost of fine-tuning LLMs in achieving optimal results, and how does it impact performance in the tourism sector?
- Q5: Can LLMs be effectively utilized for the evaluation of customer reviews, and how can they revolutionize the hospitality sector by automating review responses and extracting actionable insights?

To fulfill the objectives of our study, we conducted an extensive review of the existing literature on sentiment analysis and classification, with a specific focus on the application of LLMs in the tourism sector in Section 2. Section 3 details the materials and methodologies utilized in this research, while Section 4 presents the prediction outcomes for all three models and their variations, both before and after fine-tuning. In Section 5, we engage in a thorough discussion of the results, extracting valuable insights and statements derived from our research findings.

2. Literature Review

2.1. The Applications of AI in Modern Hospitality

The hospitality and tourism sectors are increasingly utilizing advanced information systems and technologies, particularly artificial intelligence (AI) and deep learning, to improve service delivery, enhance customer experiences, support decision-making, and predict market trends. This section synthesizes recent research that explores the interplay between sentiment analysis, AI applications, and tourism demand forecasting, shedding light on the methodologies employed and the implications for industry stakeholders [13].

2.1.1. Sentiment Analysis on Customer Reviews in the Tourism Sector

Sentiment analysis has become vital for extracting valuable insights from unstructured customer reviews. Priya et al. (2023) [14] highlight the challenges of noisy data, such as spelling errors and emoticons, in analyzing hotel reviews. Combining fine-tuned Bidirectional Encoder Representations from transformers (BERT) with recurrent neural networks (RNNs), their proposed model significantly enhances sentiment classification accuracy. This approach demonstrates the efficacy of advanced neural networks and underscores the necessity of robust models to handle real-world data complexities.

Similarly, Wen et al. (2022) [15] conducted a comparative analysis of the BERT and ERNIE models, revealing that, while BERT effectively categorizes sentiments, ERNIE outperforms it in both accuracy and stability. This study emphasizes the importance of selecting appropriate models for sentiment analysis in understanding customer preferences, which is crucial for improving hotel recommendation systems.

Zheng et al. (2009) [3] address the need for sentiment analysis of Chinese traveler reviews, a relatively underexplored area compared to English-language reviews. Using the SVM algorithm, the authors analyzed hotel reviews from Ctrip.com to classify sentiments and demonstrate that SVM can achieve high performance for Chinese reviews. Their work highlights the importance of adapting sentiment analysis methods to specific languages and cultures, with implications for the tourism industry in China, the world's largest internet user base.

Wadhe et al. (2020) [5] take a more technical approach, applying ML techniques to classify tourist place reviews. Their study compares various feature extraction methods, such as CountVectorization and TFIDFVectorization, and evaluates classification algorithms such as Naive Bayes, SVM, and RF. Their findings indicate that TFIDFVectorization, when combined with RF, delivers the highest accuracy in sentiment classification, providing a valuable methodology for analyzing large volumes of user-generated content in the tourism industry.

Kumar et al. (2024) [6] introduce an aspect-based sentiment analysis approach to predict travel destination ratings, using Multinomial Logistic Regression and a fuzzy domain ontology algorithm. Their work is focused on TripAdvisor reviews, aiming to predict user sentiment more accurately by considering aspects such as weather conditions, service quality, and location. By incorporating dimensionality reduction through Random Projection and leveraging GloVe word vector representations, the authors achieved high classification accuracy, highlighting the potential for personalized travel recommendations based on user sentiment analysis.

Martín et al. (2018) [7] contribute to the growing field of sentiment analysis by applying deep learning techniques to classify tourist sentiments in online reviews. Their work compares classifiers based on CNN and LSTM networks, using data from Booking.com and TripAdvisor. They demonstrate that LSTM-based models are particularly effective in capturing nuanced sentiments in reviews, offering a promising direction for improving sentiment classification accuracy and enhancing customer satisfaction predictions in the tourism industry.

Li et al. (2018) [8] propose a novel Bidirectional Recurrent Neural Network (BiGRULA) model enhanced by an attention mechanism and topic-enriched word vectors (lda2vec). Their approach addresses the limitations of traditional sentiment analysis models by considering both the context of words and the topics discussed in reviews. The model outperforms existing methods in sentiment classification, providing a more coherent understanding of tourist sentiments and offering a more effective tool for analyzing hotel reviews and other tourism-related data.

Kusumaningrum et al. (2023) [16] further contribute to this discourse by developing Sentilytics 1.0, a web-based application that employs convolutional neural networks (CNNs) and improved long short-term memory (LSTM) models to analyze multilevel sentiment in Indonesian hotel reviews. The application's performance metrics indicate its robustness, and it provides actionable insights for hotel management through comprehensive sentiment visualization.

Chang et al. (2023) [17] take a different approach by proposing a heuristic model for sentiment analysis in luxury hotels. They identify key features impacting customer satisfaction by integrating visual and multimedia analytics. Their findings suggest that luxury hotels should focus on staff training and location selection to enhance customer experiences, reflecting the nuanced understanding of customer sentiments.

2.1.2. AI and Demand Forecasting

The integration of sentiment analysis into decision-making and demand forecasting is explored by Zhang et al. (2024) [18], who introduce the long short-term memory interaction-based convolutional neural network (LICNN) framework. Their study demonstrates significant improvements in forecasting accuracy when online review features are incorporated, suggesting that customer sentiment can be a valuable predictor of hotel demand. This innovative approach bridges the gap between sentiment analysis and operational forecasting, providing hotel managers with actionable insights for strategic planning.

Ounacer et al. (2023) [19] emphasize the significance of aspect-based sentiment analysis in tourism, presenting a semi-supervised CorEx method for aspect extraction. Their findings highlight the necessity of nuanced sentiment analysis in aiding customer decision-making, aligning with the broader trend of leveraging customer feedback to inform business strategies.

2.1.3. AI Applications in Service Delivery

The role of AI in enhancing service encounters within the hospitality sector is systematically reviewed by Li et al. (2021) [20], who identify four modes of AI interactions and their impact on customer service outcomes. Their model provides a framework for understanding how AI can improve service delivery, particularly during public health emergencies such as COVID-19.

Pillai et al. (2020) [21] expand on this by investigating customer behavioral intentions toward AI chatbots in the Indian hospitality sector. Their mixed-methods approach reveals that perceived ease of use, trust, and anthropomorphism significantly influence chatbot adoption intentions, offering practical insights for designers and managers aiming to enhance AI integration in travel planning.

Huang et al. (2022) [22] propose a comprehensive evaluation framework for AI adoption in hospitality, identifying high-adoption areas such as search engines and virtual agents. This qualitative synthesis elucidates the factors influencing AI adoption, facilitating strategic decision-making.

2.1.4. Advanced Technologies and Tourism Experiences

Recent studies also explore the transformative potential of emerging technologies in enhancing tourist experiences. Miao et al. (2023) [23] examine generative AI tools and their implications for marketing and engagement in tourism. Their research highlights how these tools can enable personalized visual experiences, enriching tourist interactions.

Wang et al. (2020) [24] discuss the intersection of IoT, 5G technology, and AI in smart tourism, suggesting that these technologies can significantly improve the quality and efficiency of tourist services. Their case study demonstrates the effectiveness of combining these technologies to create innovative solutions for enhancing customer experiences.

2.1.5. Challenges and Future Directions

Despite promising advancements in AI and sentiment analysis, challenges remain in achieving seamless integration and addressing user privacy concerns. Chi et al. (2022) [25] and Gupta et al. (2023) [26] explore tourists' attitudes toward AI in service delivery and the role of facial recognition technology, respectively. Their findings indicate that acceptance varies across contexts and highlight the need for careful consideration in service design.

Furthermore, Zhang et al. (2023) [27] assess the quality of AI-generated content compared to human-generated content, advocating for human-machine collaboration to enhance the nuanced quality of digital tourism interpretation.

The literature reviewed in this section underscores the critical role of sentiment analysis and AI technologies in transforming the hospitality and tourism sectors. By harnessing customer insights and advanced data processing techniques, businesses can enhance service delivery, improve customer satisfaction, and forecast demand more accurately.

2.2. Application of LLMs in the Tourism Sector

After discussing the broader impact of AI on the tourism sector, we turn our attention to the specific advancements in LLMs. Their rapid developments have revolutionized the industry by transforming the way information is managed and generated by information systems, recommendations are personalized, and tourism services are optimized. Numerous studies have investigated the potential of LLMs to enhance the creation of tourism-related information and deliver tailored services. This highlights the significant role LLMs play in improving the tourist experience, refining recommendation systems, and addressing sustainability and service quality challenges.

2.2.1. Context-Specific Applications

Qi et al. (2024) [10] investigate the limitations of existing smart tourism systems in Tibet, emphasizing the difficulties in generating culturally and contextually relevant content. They introduce the DualGen Bridge AI system, which employs supervised fine-tuning to provide personalized descriptions of tourist sites, highlighting the need for region-specific LLM applications to capture cultural nuances effectively. Similarly, Wei et al. (2024) [28] present TourLLM, a model fine-tuned with a specialized dataset, Cultour, to improve travel-related information quality. They introduce the CRA (consistency, readability, availability) criterion for evaluating LLM-generated content, demonstrating that domain-specific fine-tuning enhances relevance and accuracy.

2.2.2. Sustainable Tourism and Recommendations

Banerjee et al. (2024) [29] explore how LLMs can enhance tourism recommender and information systems (TRSs) by integrating sustainability metrics into the recommendation process through a sustainability augmented re-ranking (SAR) model. Their findings indicate that LLM-enhanced systems outperform traditional models, particularly when incorporating sustainability considerations. Vasic et al. (2024) [30] expand on this by examining LLMs' role in creating personalized museum tours, showing how tailored experiences can significantly enhance user satisfaction.

2.2.3. Advancements in Recommender Systems

Chen et al. (2023) [31] discuss the evolution of recommender and information systems, noting a shift toward conversational systems empowered by LLMs. They highlight LLMs' superior memory and reasoning capabilities, which streamline operations and enhance user experience. In line with this, Balamurali et al. (2023) [32] and Falatouri et al. (2024) [33] examine LLMs' integration with sentiment analysis and service quality assessment. Their research underscores the ability of LLMs to provide real-time, emotionally attuned responses, significantly enhancing customer experience and service quality.

2.2.4. Combining Human and AI Insights

Yñiguez-Ovando et al. (2024) [34] propose a hybrid framework that combines LLMs with human intelligence to address sustainability issues, particularly over-tourism. This approach illustrates the potential for LLMs, when paired with human insights, to formulate effective solutions. Secchi et al. (2023) [35] investigate the integration of LLMs with tourism-specific knowledge graphs, enhancing the quality of hospitality services through improved information retrieval.

2.2.5. Tourism Promotion and AI Chatbots

Kodors et al. (2024) [36] emphasize using LLMs in AI-driven chatbots for tourism promotion, focusing on personalized, context-aware interactions with tourists. Their findings highlight trustworthiness and information saturation as critical factors for effective chatbot design. Hsu et al. (2024) [37] address limitations in generic LLMs and advocate for multistakeholder datasets to improve the reliability of generated tourism information.

2.2.6. Challenges and Ethical Considerations

Despite the advancements, challenges remain. Qi et al. (2024) [38] note the issue of hallucinations in LLM-generated content, proposing an RAG-optimized model to mitigate inaccuracies. Balfroid et al. (2024) [39] and Gonzalez-Garcia et al. (2024) [40] highlight additional limitations in automated content generation and knowledge graph enhancement. Meyer et al. (2024) [41] compare LLM fine-tuning methods, stressing the necessity for human oversight and robust evaluation metrics to ensure effective chatbot performance.

2.2.7. Potential Risks and Future Research Directions

Carvalho et al. (2024) [42] offer a balanced analysis of LLMs' benefits and risks, discussing applications in customer service and itinerary planning while raising concerns about job displacement and misinformation. Sioziou et al. (2024) [43] explore LLMs' capabilities in extracting structured information from the tourism job market, indicating potential efficiencies in recruitment processes. Finally, Liyanage et al. (2023) [44] address the growing concern of AI-generated opinion spam in hotel reviews, revealing the challenges in maintaining trust in online review systems.

In summary, the literature underscores the transformative potential of LLMs in the tourism sector while also addressing ongoing challenges such as content accuracy, sustainability, and ethical considerations.

3. Materials and Methods

In the literature review section, previous studies on adopting AI and LLMs in tourism were presented. These studies collectively highlighted the wide range of AI applications that support the tourism sector in its highly competitive landscape.

This study focuses on utilizing the advanced capabilities of LLMs, specifically the latest GPT omni family and the widely researched BERT natural language processing model. The GPT omni family consists of two separate models: GPT-4o, which is the high-intelligence OpenAI's flagship model designed for complex and multi-step tasks, and GPT-4o mini, which is more affordable and optimized for faster, more lightweight tasks. Both GPT models are more capable than their predecessors, GPT-4 turbo and GPT-3.5 turbo, generating text twice as fast at half the cost [12].

The main objective of our research is to stress-test these three models with a demanding classification task to determine if they can deliver real-world results that support the tourism sector.

In the first phase of our research, we carefully pre-processed and cleaned our dataset. We then prompted the GPT base models to predict the star ratings for each review in our dataset (zero-shot). Later, we fine-tuned BERT and both GPT omni models using few-shot learning and parameter tuning, prompting them again to make predictions on the same dataset. Finally, we conducted a direct comparison, extracting valuable insights from the results.

3.1. Dataset Cleaning, Preprocessing, and Splitting

Tripadvisor is a well-known online travel platform that offers travelers a wealth of information, reviews, and recommendations on hotels, restaurants, attractions, and other travel-related services worldwide. It allows users to plan and book trips by exploring user-generated content, such as ratings, photos, and in-depth reviews, while also providing the ability to compare prices for accommodations, flights, and activities.

With its extensive collection of verified user-generated content, Tripadvisor stands out as one of the most reliable sources for tourism-related reviews, often considered more trustworthy than platforms such as Google My Business or Google Maps.

Recognizing its credibility, we sourced the "Trip Advisor Hotel Reviews" dataset from the Kaggle platform, which includes 20,491 customer reviews, specifically from hotels. This dataset, with a total size of 5 MB, has been highly rated by users, earning a usability score of 10 for its completeness, credibility, and compatibility. It is available to the public under the Attribution-NonCommercial 4.0 International license, which allows for copying and redistribution of the material in any medium or format for non-commercial use, as long as appropriate credit is given and a link to the license is provided [45,46].

3.1.1. Dataset Analysis

To better understand the dataset's representativeness and potential biases, we conducted a detailed analysis of its characteristics.

Rating Distribution

The dataset contains a range of ratings from 1 to 5 stars. The 1-star reviews account for 1421 entries, making up 6.93% of the total dataset, while the 2-star reviews represent 1793 entries, or 8.75%. The 3-star reviews contribute 2184 entries, which is 10.66% of the dataset. The largest portions of reviews come from higher ratings, with 6039 4-star reviews (29.46%) and 9054 5-star reviews (44.20%).

As seen in Figure 1, there is a noticeable skew toward higher ratings, with 73.66% of reviews being 4-star or 5-star. This imbalance reflects a common trend in user-generated content, where satisfied customers are more likely to leave feedback. However, this also highlights the need to carefully interpret sentiment analysis results, as the dataset might not fully represent the entire spectrum of customer experiences.

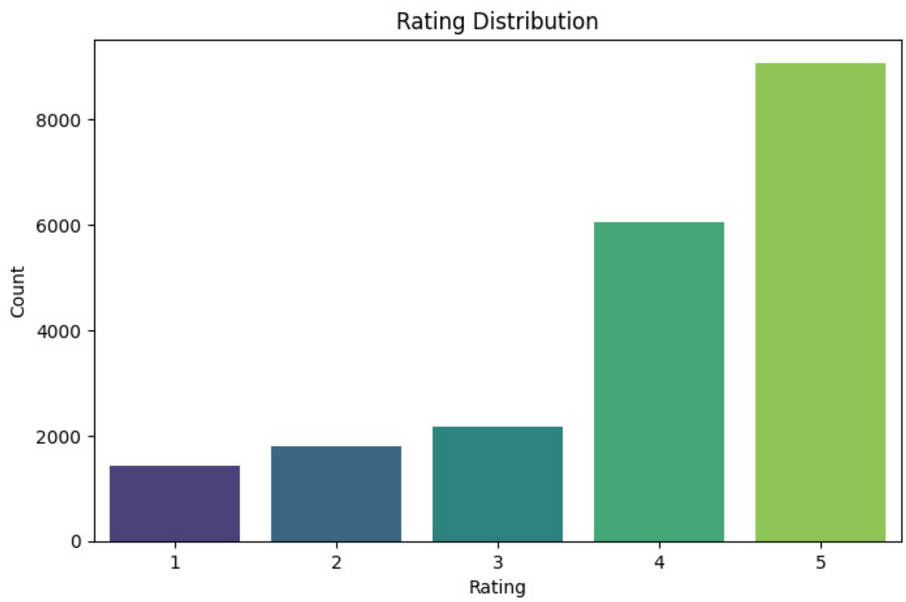

Figure 1. Rating Distribution.

Sentiment Mapping and Distribution

To facilitate sentiment analysis, we mapped the numerical ratings to three sentiment categories: positive, neutral, and negative. Specifically, reviews with ratings of 4 and 5 were classified as positive, comprising 73.66% of the dataset. A rating of 3 was assigned to the neutral category, representing 10.66% of the reviews, while ratings of 1 and 2 were categorized as negative, accounting for 15.68% of the dataset.

As expected, the sentiment distribution follows the overall rating distribution, with a large proportion of positive sentiments. As we mentioned earlier, this mirrors common trends in user-generated content, where satisfied customers are more likely to leave feedback. However, this imbalance in the dataset presents a potential challenge for training machine learning models. If we were to train sentiment analysis models on this dataset as it stands, the models would likely become biased toward predicting positive sentiments due to the overwhelming number of 4-star and 5-star reviews.

To address this, further data preprocessing and careful splitting of the dataset would be necessary to ensure a more balanced distribution across sentiment categories. Techniques such as oversampling the underrepresented negative and neutral categories or undersampling the dominant positive category could help create a more equitable dataset. This approach would improve the model's ability to generalize and accurately classify all types of reviews, including those with negative or neutral sentiments, leading to more reliable and fair sentiment analysis outcomes.

Review Length Analysis

The review length also plays a vital role in training machine learning models, as the length of text can impact the quality of the features extracted for sentiment analysis and other tasks. In our dataset, the average review length is approximately 725 characters, with a wide range from 44 to 13,501 characters. The shortest review is 44 characters long, while the longest extends to 13,501 characters. The median review length is 537 characters, suggesting that most reviews are relatively concise. In fact, 50% of the reviews fall within a range of 339 to 859 characters. This variation indicates a diversity in the level of detail provided by users, from brief comments to more comprehensive feedback. However, the presence of outliers, particularly extremely long reviews, could affect text-based analysis. To mitigate the impact of these outliers, preprocessing techniques such as truncation or weighting might be necessary to ensure that the model is not disproportionately influenced by overly long reviews.

Word Cloud Analysis

To explore the diversity of vocabulary within the reviews, we generated a word cloud that highlights the most frequently used terms (Figure 2). Common words include "hotel", "room", "time", and "night", which are consistent with the primary topics customers emphasize in hospitality reviews. This suggests that the dataset captures relevant and context-specific language, adding value to sentiment and emotion analysis tasks.

Figure 2. Word cloud for user-generated content.

Dataset Limitations and Relevance

While the dataset is comprehensive and well-suited for our study, several limitations are noted:

- Rating imbalance: the predominance of positive reviews (73.66%) could introduce bias in the classification of sentiment, potentially making models overly optimized for positive feedback.
- Lack of metadata: the dataset does not include information about reviewer demographics, geographic location, or other contextual factors that might influence sentiment.
- Platform-specific bias: as the data are sourced exclusively from Tripadvisor, they may not fully represent reviews from users who prefer other platforms, such as Yelp or Booking.com.

Despite these limitations, the dataset remains highly relevant for our objectives. It provides a realistic benchmark for evaluating advanced language models in processing and interpreting user-generated content, which is a critical task in the tourism and hospitality industries.

3.1.2. Dataset Preprocessing

To ensure the quality and effectiveness of predictive modeling and fine-tuning, we prepared the dataset through a structured approach. The dataset initially included two columns, "Review" and "Rating", and an additional "ID" column was added to help trace potential errors. The "Rating" column, serving as the label for model predictions, consisted of integer values from 1 to 5, as specified by the dataset's author.

Preprocessing focused on refining the "Review" column, which involved removing special characters, extraneous whitespace, and empty rows to enhance data consistency and reduce noise. The text normalization steps included converting accented characters to base forms and transforming all text to lowercase, ensuring uniformity and improving model performance [47].

3.1.3. Dataset Splitting

As highlighted in the dataset analysis in Section 3.1.1, there is a marked difference in the frequency of each label, indicating that the dataset is imbalanced, with a significant overrepresentation of higher ratings (5 and 4) in contrast to the lower ratings (1, 2, and 3). This label imbalance can introduce bias into the model, causing it to become overly tuned to the majority classes (5 and 4) while struggling to accurately predict the minority classes (1, 2, and 3). Training a model on such skewed data may lead to overfitting, whereby the model excels at predicting the majority classes but fails to generalize effectively to the minority classes. As a result, this could adversely impact predictive performance, particularly for the lower ratings.

The goal is to create a balanced subset of 5000 reviews, with an equal number of samples distributed across five labels (review ratings 1–5). To achieve this, we applied an undersampling technique to address the class imbalance in the original dataset. By ensuring the equal representation of each class, undersampling helps prevent the model from becoming biased toward the majority classes and improves its ability to generalize across all labels. We used the StratifiedShuffleSplit method to preserve the original class proportions within the subset while maintaining randomness in the selection process. Additionally, we set a random_state of 42 to guarantee the reproducibility of the results, ensuring that our approach can be independently validated and yielding consistent subsets across multiple runs.

The dataset was then divided into training, testing, and validation sets using the same approach, with the stratified column passed as a parameter in the train_test_split function. Initially, we split the data into 80% training and 20% testing sets. The training set was then split further, allocating 20% for validation and 80% for further training. Throughout fine-tuning, the training set was essential in helping the model learn patterns and relationships within the data, enabling it to perform tasks and make accurate predictions. The validation set was used to optimize hyperparameters and model configurations. Monitoring the performance on this set allowed for adjustments that improved generalization and reduced overfitting.

After splitting the dataset, we obtained a test set of 1000 samples, a training set of 3200 samples, and a validation set of 800 samples, all equally distributed across the five labels.

3.2. LLM Prompt Engineering

Our aim was to create a prompt that seamlessly integrates with various LLMs, enhancing the usability of their outputs through our code. We prioritized not only the prompt's content but also how its output is presented for improved accessibility.

To develop a prompt compatible with multiple LLMs, we needed a deep understanding of the distinct models available, such as GPT, Claude, and LLaMA, each with its unique strengths and limitations. Crafting a prompt that could generate coherent responses from all these models, while remaining user-friendly, posed a significant challenge. To address this, we employed two key prompt engineering strategies, tailored to the specific traits of each LLM [48].

- Model-independent content: We designed the prompt to be agnostic of any particular model's architecture or nuances. This adaptability ensures that the prompt can be applied across different LLMs with minimal adjustment. The focus was on constructing a prompt that clearly communicates the task, providing relevant context and information that can be interpreted by any LLM.
- Output formatting for accessibility: Recognizing the importance of usable output, we emphasized creating a structure that aligns well with coding and accessibility needs. This required organizing the responses in a logical, intuitive manner, specifically formatting the output to comply with the JSON standard [49].

After several iterations and tests with various LLMs, we finalized a prompt that consistently produces outputs in the desired format that are easily interpreted by both humans and models, as shown in Listing 1.

Listing 1. Model-agnostic prompt.

```
conversation.append({'role': 'system',
'content': "You are an AI model tasked with predicting a star rating (1–5) based on the user's review"
        "of their hotel experience. Return your response in JSON format: {'rating': integer}."})
conversation.append({'role': 'user',
'content': f"Predict the star rating (integer between 1 and 5) to the following review. Return your response"
        f"in JSON format similar to this example {{'rating': integer}}. Please avoid providing additional"
        explanations. Review:\n{input['Review']}"})
```

3.3. Model Deployment, Fine-Tuning, and Predictive Evaluation

The aim of this study is to determine which of our three models performs best in predicting users' ratings based on the content of their reviews, specifically for sentiment analysis and classification. By identifying the model that most effectively understands the feelings and concerns of hotel customers, we can develop a powerful tool for automatically gaining insights from reviews. This would enable businesses to make informed decisions that enhance customer satisfaction, ultimately driving better outcomes and improving overall success.

To achieve this, all three models were tasked with making predictions on the test set, both before and after fine-tuning using few-shot learning. To ensure fairness during training, we applied the same hyperparameters across all models: a learning rate of 2×10^{-5}, a batch size of 6, and training over 3 epochs, using the Adam optimizer during fine-tuning. The selected hyperparameters for fine-tuning were chosen to balance optimization efficiency and computational feasibility while ensuring consistent evaluation across the GPT-4o, GPT-4o mini, and BERT models. A learning rate of 2×10^{-5} is commonly used for fine-tuning large pre-trained language models, as it effectively avoids the risks of overshooting optimal parameter settings (with higher rates) or slow convergence (with lower rates).

The batch size of 6 reflects a compromise between computational resource constraints and the need for stable gradient updates, particularly given the model sizes and GPU memory limitations. Training for 3 epochs was empirically determined to provide sufficient updates for meaningful fine-tuning without overfitting, which was validated by monitoring validation loss during training. The Adam optimizer was employed for its adaptive learning rate capabilities, ensuring robust and efficient optimization across diverse model architectures.

The choice of these hyperparameters for fine-tuning LLMs such as GPT-4o and GPT-4o mini often follows default settings provided by OpenAI's API [50], which are designed to prevent overfitting while ensuring efficient learning. OpenAI's documentation recommends smaller batch sizes and fewer epochs, especially for use cases with limited data, to achieve optimal results without extensive experimentation. These configurations are informed by prior research and widespread application performance, making them a practical and effective standard [47,51]. By maintaining consistency across all models, this approach ensures fair evaluation while leveraging configurations that are well-suited to the characteristics of transformer-based architectures.

Although it is beyond the scope of this research, it is important to note that for production use, the fine-tuning process should involve extensive experimentation with different hyperparameter settings to identify the most effective and efficient configuration. Techniques such as Grid Search [52], Random Search [53], and advanced methods such as Bayesian Optimization or Tree-structured Parzen Estimators (TPEs) [54] can be employed to optimize performance while balancing cost-effectiveness. However, it should be emphasized that applying hyperparameter optimization (HPO) techniques to LLMs fine-tuned through APIs can incur substantial computational costs. These costs stem from the need to repeatedly fine-tune and evaluate the model under various configurations, which is resource-intensive given the scale of LLMs.

The BERT model was configured to handle inputs of up to 512 tokens (around 2560 characters). Reviews exceeding this length were truncated to fit within the token limit. While methods such as chunking or hierarchical processing could manage longer texts, they introduce additional complexities. Similarly, the GPT models have a token limit of approximately 16,384, requiring truncation for longer reviews during prediction.

Below, we outline the deployment strategy for each model, detailing our unique approach in applying them to the tasks of sentiment analysis, spam detection, and classification.

3.3.1. GPT Model Deployment and Fine-Tuning

In this phase, we used the GPT omni family, specifically the GPT-4o and GPT-4o mini models, both in their base forms and after fine-tuning, to classify customer reviews into the appropriate rating label (star rating). Initially, using the prompt shown in Listing 1, we applied the base models without any additional training (zero-shot) to predict the ratings for the test set. Thanks to their extensive pre-training on large datasets, these GPT models are capable of making fairly accurate predictions even without fine-tuning.

To facilitate communication between our software and the GPT models, we utilized OpenAI's official API. This allowed us to send prompts containing the review text from the feature column and receive the model's predictions in JSON format. The predictions from both models were then stored in two separate columns within the same test_set.csv file.

During the fine-tuning phase, the same models underwent additional training to boost their performance by learning from prompt-completion pairs in our training dataset. This fine-tuning process helped the models to better understand the subtle nuances and patterns within the data. By employing a multi-epoch training approach, the models continuously improved their comprehension and capabilities through multiple iterations, leading to more accurate predictions and enhanced task execution. This iterative process ensured that the models were well-equipped to provide precise predictions and valuable insights during the prediction phase.

To proceed with the fine-tuning process of our models, we generated two JSONL files containing pairs of prompts and their corresponding completions, as shown in Listing 2.

Listing 2. Prompt and completion pairs–JSONL files.

```
{"messages": [{"role": "system",
        "content": "You are an AI model tasked with predicting a star rating (1–5) based on the
        user's review of their hotel experience. Return your response in JSON format: {'rating':
        integer}."},
        {"role": "user", "content": "..."},
        {"role": "assistant", "content": "{'rating': 2}"}]}
```

For our study, we initiated two fine-tuning tasks by uploading our training and validation JSONL files into OpenAI's user interface. These tasks were identified by the job IDs: ftjob-J0ApzOhA7HqE1UxBp7UBAm7U and ftjob-pDcZW0NmELoyiotCSuNjnzI1.

The first fine-tuning process focused on the GPT-4o model, which was trained on 2,167,176 tokens. The initial training loss started at 0.4292 and decreased to 0.0208, with a validation loss of 0.1719. The second fine-tuning process involved the GPT-4o mini model, also trained on 2,167,176 tokens. This model began with an initial training loss of 0.5228 and reduced it to 0.0218, achieving a validation loss of 0.1131. These metrics highlight the models' impressive effectiveness for this task and its ability to generalize during the fine-tuning process.

After completing the fine-tuning, both models were used to make predictions on the same test set as their base versions, with the results stored in separate columns.

3.3.2. BERT Model Deployment and Fine-Tuning

In this phase, we focused on training the BERT model, specifically the bert-base-uncased variant [55], for the same classification task addressed by the LLMs. Using the BertForSequenceClassification model from the transformers' library, we leveraged BERT's transformer architecture to generate contextual token representations and predict class probabilities via its classification head. This variant of BERT features 12 layers, 768 hidden units, 12 attention heads, and 110 million parameters, with self-attention mechanisms capturing contextual dependencies [55].

The fine-tuning phase involved rating prediction classification, pairing reviews with their corresponding labels. Conducted on Google Colab with a Tesla V100-SXM2-16GB GPU (NVIDIA, Santa Clara, CA, USA), the process included tokenization using the BERT tokenizer, hyperparameter adjustments, and optimization via the Adam optimizer. The model was trained over three epochs, with progress tracked using the tqdm library, while validation utilized the same set as the GPT-4o models. After fine-tuning, predictions were generated for the test set previously used by GPT-4o. All relevant code, metrics, and validation results are available in an IPython notebook hosted on GitHub [56].

4. Results

In Section 3, we examined the methodology used to run and fine-tune the LLMs and NLP models, detailing how these models were utilized to predict user ratings based on a test set of hotel reviews. This section presents a comparative analysis of the three models, showcasing their evaluation metrics both before and after fine-tuning.

4.1. Overview of Fine-Tuning Metrics

During the fine-tuning process, we collected crucial metrics for each model, including training loss, validation loss, training time, and training cost, all of which are presented in Table 1.

Table 1. Fine-tuning metrics.

Model	Resources	Training Loss	Validation Loss	Training Time (Seconds)	Training Cost
ft:gpt-4o	API	0.0208	0.1719	3412	$29.18
ft:gpt-4o-mini	API	0.0218	0.1131	2143	$1.38
ft:bert-adam	Tesla V100-SXM2-16 GB	0.6378	0.9526	272.30	$0.79
ft:bert-adamw	Tesla V100-SXM2-16 GB	0.6401	0.9866	278.09	$0.82

Training loss reflects the model's performance during the training phase by measuring the discrepancy between its predictions and the actual target values (e.g., user ratings). A lower training loss indicates that the model is effectively learning from the training data.

On the other hand, validation loss assesses the model's performance on a distinct validation dataset that was not used during training (validation_set). This metric is vital for determining the model's ability to generalize. Ideally, validation loss should decline as the model improves; however, if it begins to rise while training loss continues to decrease, this could signal overfitting [57].

However, it is important to note that directly comparing validation and training losses between fine-tuned NLP models and LLMs may not be straightforward due to differences in their architectures. In contrast, we can safely compare the validation and training losses for similar architectures, such as GPT-4o and GPT-4o mini, because they share underlying structural similarities and design principles. These models operate on the same foundational architecture, allowing a more meaningful comparison of their performance metrics. Since they are variations of the same model, the differences in their training and validation losses are likely attributed to the scale of the models rather than fundamental architectural discrepancies. This commonality enables us to draw more accurate conclusions about their relative performance in similar tasks.

4.2. Model Evaluation Phase

Before presenting our results, it is important to emphasize the value of model evaluation. In machine learning and NLP, evaluating models helps us understand their performance, make informed decisions, and improve fine-tuning for specific tasks. Table 2 summarizes the evaluations for each model, including critical metrics such as accuracy, recall, and f1-score.

Table 2. Comparison of model performance metrics.

Model	Accuracy	Precision	Recall	F1
base:gpt-4o	0.662	0.6616	0.662	0.6553
base:gpt-4o-mini	0.632	0.626	0.632	0.6256
ft:gpt-4o	0.67	0.68	0.67	0.6729
ft:gpt-4o-mini	0.649	0.6611	0.649	0.6532
ft:bert-adam	0.606	0.6189	0.606	0.6079
ft:bert-adamw	0.59	0.6004	0.59	0.5932

4.2.1. Pre-Fine-Tuning Evaluation

In the initial phase of our study, we deployed the GPT-4o and GPT-4o mini models to assess the capabilities of LLMs, particularly within the GPT omni family, for sentiment analysis and classification tasks. The results, summarized in Table 2, were promising, with the base models achieving accuracies of 66.2% for GPT-4o and 63.2% for GPY-4o mini, indicating that the mini model performed slightly less efficiently—by 3%—compared to its larger counterpart. Notably, these models achieved these accuracy levels without any specific fine-tuning, which is impressive considering that, in a rating system with a scale of 1 to 5, the model had five potential choices. If the model were to select an option at random, the probability of choosing the correct answer would be merely 20%. This highlights the comparative advantage of LLMs, which demonstrate significant accuracy in sentiment analysis and classification tasks, even without fine-tuning.

Another noteworthy observation is the comparison between the results of the base LLM models and those of the fine-tuned BERT models. The fine-tuned BERT models, specifically ft:bert-adam and ft:bert-adamw, were trained on a designated dataset to make predictions on a specific test set. However, their prediction accuracy was considerably much lower (5.6%) than that of the non-fine-tuned LLMs. This finding suggests that base models of LLMs have reached a level of capability that can surpass even specifically trained NLP models. This is not surprising, considering that the base models of LLMs are pre-trained on a wide range of tasks, continuously enhancing their performance in diverse applications.

4.2.2. Post-Fine-Tuning Evaluation

After fine-tuning, the models were deployed and prompted to predict the rating labels on the same test set. The results indicated an accuracy of 67% for ft:GPT-4o, 64.9% for ft:GPT-4o mini, and 60.6% and 59% for ft:bert-adam and ft:bert-adamw, respectively.

One notable observation is the performance disparity between the base GPT models and their fine-tuned counterparts. In the case of the fine-tuned models, ft:GPT-4o demonstrated an improvement of just 0.8% over its base model, while ft:GPT-4o mini showed a more significant enhancement of 1.7% compared to the mini base model. Although both GPT omni models exhibited performance gains, the improvement in ft:GPT-4o mini was approximately double that of its larger counterpart. A possible explanation for this could be related to the number of parameters each model is trained on. The mini model, with fewer parameters (about 8 billion [58]), benefits from a quicker fine-tuning process that requires fewer training data. Conversely, the larger model (about 1.8 trillion) may need more extensive training data to achieve greater results.

On the other hand, while the fine-tuned BERT models produced respectable results, they did not perform as effectively as the LLMs during fine-tuning. A potential factor for this discrepancy could be the hyperparameters used. As noted at the beginning of Section 3, all models were trained using specific hyperparameters (a learning rate of 2×10^{-5}, a batch size of 6, and training over 3 epochs) to ensure more objective results. Experimenting with various hyperparameters is crucial for NLP and LLMs to maximize each model's performance. Thus, it is evident that all models could achieve better performance with different hyperparameter configurations.

In addition, to reinforce our findings, we created scatterplots for each model, comparing the predictions with the labeled column. The efficiency of each model is illustrated in Figure 3. The fine-tuned GPT-4o model demonstrates the lowest MAE of 0.341, significantly outperforming both the GPT-4o mini (0.363) and BERT (0.444) models. A lower MAE indicates higher accuracy, as it reflects the model's ability to minimize the absolute error between the predicted and actual values.

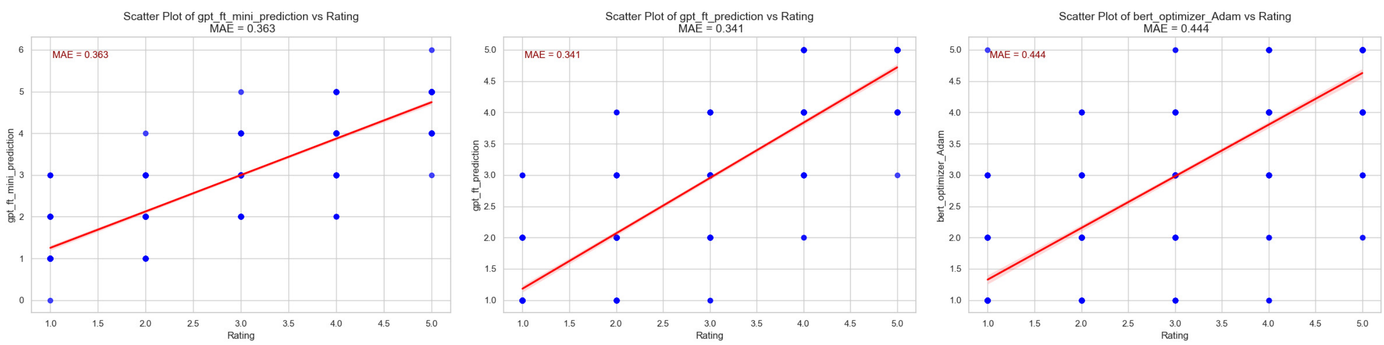

Figure 3. Scatterplots comparing MAE for fine-tuned models.

4.3. Top Influential Words Across Models

To further support our findings and provide a deeper understanding of how each model interprets the relationships between words within reviews, we applied a top influential word analysis on our test set. This technique involved identifying the most impactful words in each review that contributed to the assigned sentiment ratings (1–5). The analysis was conducted consistently across all fine-tuned models, ensuring the comparability of results and revealing insights into the differing linguistic patterns and focal points of each model.

For the sake of brevity, Listing 3 presents the top influential words for each model, focusing solely on rating 3. This comparison highlights the key words each model associates with neutral or mixed sentiment, providing insight into their distinct interpretative approaches.

Listing 3. Top influential words for rating 3 across models.

Fine-Tuned GPT-4o: [('hotel', 481), ('room', 440), ('good', 236), ('nt', 205), ('nice', 162), ('night', 139), ('stay', 123), ('great', 120), ('staff', 113), ('day', 112)]
Fine-Tuned GPT-4o-mini: [('hotel', 491), ('room', 481), ('nt', 255), ('good', 239), ('nice', 164), ('night', 155), ('great', 144), ('stay', 130), ('day', 130), ('beach', 128)]
Fine-Tuned BERT Adam Optimizer: [('room', 546), ('hotel', 467), ('nt', 262), ('good', 252), ('night', 183), ('nice', 156), ('great', 146), ('stay', 136), ('day', 135), ('staff', 132)]

The analysis of top influential words across labels reveals notable patterns in how each model processes user reviews and assigns sentiment labels. Across all models, words such as 'hotel', 'room', and 'staff' consistently appeared as key indicators of sentiment, reflecting their centrality in the context of hospitality reviews. However, fine-tuned GPT-4o models tend to place greater emphasis on adjectives such as 'great', 'good', and 'nice', which are directly indicative of sentiment, especially for higher labels (4 and 5). In contrast, BERT shows a broader focus on contextual nouns and verbs, such as 'service', 'stay', and 'desk', suggesting its stronger reliance on specific aspects of the review rather than overall tone. Additionally, GPT-4o demonstrates a more refined sensitivity to rarer and nuanced words such as 'beach' and 'location', potentially due to its larger training corpus and enhanced contextual understanding. These differences underline how GPT-4o models excel at capturing sentiment nuances, while BERT's emphasis on specific review elements may make it better suited for tasks requiring granular aspect-level analysis. These insights not only highlight the distinct strengths of each model but also emphasize the importance of aligning model selection with the specific needs of sentiment analysis tasks in the hospitality industry.

5. Discussion

In the previous sections, we employed three distinct AI models to assess their predictive capabilities on a crucial classification task: sentiment analysis of hotel customer reviews. Initially, the base models were tasked with predicting a rating [1–5] based solely on the content of the reviews. Following this, the models were fine-tuned on a specialized training set using few-shot learning and parameter-efficient tuning techniques. After fine-tuning, the models were evaluated on the same test set to generate predictions, and their performance was thoroughly analyzed.

In this section, we will leverage the classification task results to answer each research question outlined in the introduction.

5.1. NLP Models vs. LLMs: Which Is Superior for Sentiment Analysis and Star Rating Prediction?

- Research question 1: Do NLP models or LLMs demonstrate superior efficacy in assessing user-generated content such as hotel customer reviews in terms of sentiment analysis and star rating prediction?
- Research Statement 1: Fine-tuned LLMs, and in our case the GPT-4o, stand out as the most effective model for this classification task, reaffirming that larger models

with more parameters are better suited for complex tasks such as sentiment analysis. However, the use of smaller models and the selection of optimizers remain critical in balancing accuracy and computational efficiency.

In Sections 4.2.1 and 4.2.2, we presented the performance of our models both before and after fine-tuning. All three models delivered solid performance, with the fine-tuned GPT-4o model emerging as the strongest, outperforming both the fine-tuned GPT-4o mini and BERT models. Specifically, the fine-tuned GPT-4o achieved a 0.8% improvement over its base version, a 2.1% advantage over GPT-4o mini, a 6.4% lead over the fine-tuned BERT model trained with the Adam optimizer and an 8% gain over the BERT model trained with AdamW.

The results clearly demonstrate that fine-tuning significantly boosts the performance of LLMs. This underscores the importance of fine-tuning for domain-specific tasks such as sentiment analysis in hotel reviews.

The following key insights can be drawn from the results:

- GPT-4o leads the pack: The fine-tuned GPT-4o model consistently outshined its counterparts—GPT-4o mini and BERT—demonstrating superior accuracy and efficiency. This suggests that larger, more sophisticated models, when fine-tuned effectively, deliver stronger predictive performance for tasks requiring nuanced language understanding.
- Smaller models offer viable alternatives: While GPT-4o mini did not perform at the same level as its larger counterpart, it still delivered respectable results. This indicates that smaller models can be effective, especially when computational resources are constrained, though some accuracy may be sacrificed.
- BERT's performance in comparison: Although the BERT models performed reasonably well, they lagged behind the GPT-4o models, particularly after fine-tuning. The fine-tuned GPT-4o significantly outperformed both versions of BERT, regardless of whether the Adam or AdamW optimizer was used.
- Impact of optimizer selection: The choice of optimizer also played a role in model performance. BERT models trained with Adam outperformed those trained with AdamW, highlighting the importance of selecting the right training strategies.

While our results demonstrate that the GPT-4 omni model outperforms BERT in terms of accuracy and overall sentiment classification, a deeper analysis of their nuanced behaviors reveals additional insights. GPT-4 omni's superior performance can be attributed to its advanced architecture and training on diverse and extensive datasets, enabling it to better capture complex language patterns, such as sarcasm, nuanced tone, and context-dependent sentiment. On the other hand, BERT, with its bidirectional transformer architecture, excels in capturing fine-grained relationships within sentences, making it particularly effective in cases requiring precise contextual understanding. However, it is limited by its reliance on pre-training and fine-tuning for specific tasks, as opposed to GPT-4 omni's robust zero-shot capabilities. These distinctions highlight how GPT-4 omni leverages advancements initially pioneered by models such as BERT while addressing their limitations, resulting in a more versatile and powerful tool for sentiment analysis in the tourism and hospitality industry. By examining these differences, our study underscores the evolution of sentiment analysis tools and their implications for practical business applications.

5.2. GPT-4 Omni Family: Pre-Fine-Tuning Performance Comparison

- Research question 2: Which model within the GPT-4 omni family exhibits superior performance before fine-tuning?
- Research statement 2: Model size significantly influences performance in zero-shot classification tasks, with the GPT-4o model exhibiting superior accuracy compared to its mini counterpart, GPT-4o mini. This finding underscores the importance of model parameters in enabling effective language understanding and predictive capabilities.

As stated in Section 4.2.1, LLMs are essentially pre-trained models that have been trained on a wide range of data sourced from the web and private repositories. Similar to how the human brain gradually assimilates knowledge from the environment, pre-trained LLMs incrementally develop their understanding through exposure to diverse datasets. This foundational knowledge is then specialized through the fine-tuning process.

In our research, we observed that the pre-training of these models was sufficient to enable zero-shot predictions on the test set, with notable accuracy. Specifically, without fine-tuning, the base GPT-4o model achieved an accuracy of 66.2% in its predictions, while the base GPT-4o mini model attained an accuracy of 63.2%.

Naturally, when comparing the performance of these pre-trained models, the GPT-4o demonstrates a 3% advantage over its mini counterpart, given that the former is trained on 1.8 trillion parameters compared to the 8 billion parameters of the mini version [58].

Additionally, it is worth noting that even without specialized training, the base models of the LLMs outperformed the BERT model—specifically designed for the classification task of our study—by 5.6% and 2.6%, respectively. This indicates the robust capability of LLMs in handling classification tasks, even in their pre-trained state.

From the findings presented, several conclusions can be drawn:

- Effectiveness of pre-training: The ability of pre-trained LLMs, such as GPT-4o and GPT-4o mini, to perform zero-shot predictions with significant accuracy demonstrates the effectiveness of their pre-training on diverse datasets. This highlights the models' inherent capabilities to generalize and apply learned knowledge to new tasks without requiring extensive retraining.
- Model size and performance: The observed 3% performance advantage of the GPT-4o model over the GPT-4o mini reinforces the idea that larger models with more parameters can capture more nuanced patterns in data, leading to improved predictive accuracy. This underscores the importance of model size in achieving superior performance in NLP tasks.
- Robustness of LLMs compared to traditional models: The fact that both base LLM models outperformed the BERT model—despite BERT being specifically trained for the classification task—indicates that modern LLMs possess a strong foundation in language understanding that allows them to excel even without specialized fine-tuning. This suggests a shift in the landscape of NLP, where pre-trained models can rival or surpass traditional models that have undergone task-specific training.
- Potential for domain-specific applications: The ability of these models to achieve respectable performance without fine-tuning opens avenues for their application in various domains, particularly in situations where computational resources for training are limited. This flexibility makes LLMs a valuable tool for real-world applications, such as sentiment analysis in customer reviews.

5.3. GPT-4 Omni Family: Post-Fine-Tuning Performance with Few-Shot Learning

- Research question 3: Which model within the GPT-4 omni family demonstrates superior performance after fine-tuning with few-shot learning?
- Research statement 3: The fine-tuned GPT-4o model exhibits overall superior performance compared to its mini counterpart; however, the mini model shows greater improvement in performance after fine-tuning, suggesting that it adapts more effectively to few-shot learning scenarios despite having fewer parameters.

In Section 4.2.2, the fine-tuned GPT-4o once again proved to be more efficient than its mini counterpart by 2.1%. However, its performance after fine-tuning was only 0.8% better, whereas the mini model improved its performance by 1.7% following fine-tuning. This specific difference in performance raises further questions: Could it be that the GPT-4o has reached a state of overfitting with the given training set? Or perhaps the GPT-4o-mini model was able to adjust its parameters more rapidly due to its smaller size?

The first possibility can be ruled out, since Table 1 shows a lower training loss for the fine-tuned GPT-4o model (0.0208 compared to 0.0218). However, if we look at the validation loss column in the same table, we may find some answers (0.1719 versus 0.1131). Thus, during fine-tuning, the mini model exhibited a lower validation loss.

The second possibility, however, seems more significant, as a model with 1.8 trillion parameters may have a harder time specializing its knowledge compared to a model with 8 billion parameters. In other words, a smaller model can better adapt its knowledge with a limited labeled dataset, while a larger model requires substantially more training data to achieve similar or superior performance.

From the provided analysis, several key findings can be derived:

- Performance comparison: The fine-tuned GPT-4o outperforms its mini counterpart by 2.1% in overall efficiency. However, the fine-tuned GPT-4o shows only a marginal improvement of 0.8% post fine-tuning, whereas the mini model demonstrates a more substantial enhancement of 1.7%. This indicates that, while the GPT-4o is more effective overall, it may have limitations in adapting further from its fine-tuning process compared to the mini model.
- Possible overfitting: The small improvement in performance of the GPT-4o after fine-tuning raises questions about potential overfitting. While the lower training loss for the fine-tuned GPT-4o suggests that it is learning effectively, the subsequent validation loss comparison indicates that the mini model is not only learning but generalizing better to unseen data.
- Validation loss insights: The lower validation loss of the GPT-4o mini model (0.1131) compared to the GPT-4o (0.1719) implies that the mini model has a better capacity for generalization during the fine-tuning phase. This suggests that the mini model, despite its smaller size, is capable of effectively adjusting its parameters to achieve a more robust performance on validation data.
- Impact of Model Size on Specialization: The analysis highlights the inherent challenges larger models face in specializing their knowledge when fine-tuning. The GPT-4o, with 1.8 trillion parameters, may require more extensive and varied training data to fine-tune effectively, while the smaller GPT-4o mini can adapt more quickly and effectively to specific tasks with limited labeled data. This suggests a trade-off between model size and adaptability in certain contexts.
- Training Data Requirements: The findings reinforce the notion that larger models need significantly more training data to achieve similar or superior performance as smaller models in few-shot or limited data scenarios. This is particularly relevant for applications where data availability is constrained.

5.4. Evaluating the Cost of Fine-Tuning LLMs: Performance vs. Expense in the Tourism Industry

- Research question 4: How significant is the cost of fine-tuning LLMs in achieving optimal results, and how does it impact performance in the tourism sector?
- Research statement 4: The cost of fine-tuning LLMs is a critical factor in achieving optimal results, as demonstrated by the significantly lower training costs and improved performance of the GPT-4o mini model compared to the larger GPT-4o, making it a more practical and economically viable option for the tourism sector, where training on large datasets can be prohibitively expensive.

In Table 1, we observe that the GPT-4o model requires 56.86 min for training, while the GPT-4o mini takes only 35.71 min, and the BERT model requires just 4.53 min. Additionally, the training costs reveal that fine-tuning the GPY-4o with 3200 training samples and 800 validation samples costs $29.18, compared to $1.38 for the mini and an even lower $0.79 for BERT. In other words, training the GPT-4o is 21.1 times more expensive than training the mini model and 36.94 times more expensive than training the BERT model. This significant difference in both time and cost between the larger GPT-4o and the smaller models is clearly due to the size, as the larger model requires considerably more time and GPU resources for fine-tuning compared to the smaller models.

However, there is a disparity in training time and costs between the GPT-4o and the GPT-4o mini. Although the fine-tuning cost of the GPT-4o is 21.1 times higher, the training time is only 1.59 times greater than that of the mini counterpart. This raises questions about how model training costs are calculated. The answer lies not in the training time but in the resources required for each model. A larger model necessitates more GPUs for training, while a smaller model requires fewer. To decouple charges based on resources and time, OpenAI and similar companies have adopted the concept of tokens (a descriptive output of 1000 characters, calculated through the tiktoken library, corresponds to 204 tokens [11]). In our case, the GPT-4o cost USD 2.50 for every 1 million input tokens and $10 for every 1 million output tokens, while the GPT-4o mini costs only USD 0.15 for each million input tokens and $0.60 for each million output tokens [59], making the mini model significantly more economical for training.

Having established which model is more cost-effective during training, we can confidently explore the performance relative to the cost. In Research Statement 3, we noted that the GPT-4o mini model, while less efficient than its larger counterpart, improved its performance by 1.7% compared to a 0.8% improvement for the GPT-4o using the same training and validation set. Therefore, when considering the performance of the GPT-4o mini in relation to its training costs, it clearly offers better value for money. Specifically, in the tourism sector, where we deal not with thousands but millions of training data, the cost of training a large model such as GPT-4o would be prohibitively expensive for hotels, whereas the mini model would be much more affordable.

The long-term cost implications of deploying LLMs such as GPT-4o in the hospitality industry must be carefully considered, as these models incur significantly higher expenses for both fine-tuning and deployment compared to more traditional models such as BERT. While GPT-4o offers superior performance in terms of understanding complex language patterns and generating nuanced sentiment analysis, its reliance on external APIs for fine-tuning and inference can lead to substantial ongoing costs. These costs are driven by the continuous usage of the model, which may require frequent calls to the API for processing customer reviews, resulting in high operational expenses, particularly for large-scale applications within the hospitality industry. In contrast, BERT, which can be fine-tuned on local infrastructure such as Tesla V100 GPUs, incurs lower costs after the initial fine-tuning phase. While the upfront costs of GPU infrastructure may be significant, the lack of ongoing external service fees for deployment makes BERT a more cost-effective option for long-term, large-scale sentiment analysis tasks. This difference in operational costs between LLMs and traditional models such as BERT underscores the importance of considering the financial sustainability of these technologies, especially in industries such as hospitality, where the volume of data and frequency of use can make a substantial impact on overall costs.

5.5. Leveraging LLMs for Customer Review Evaluation: Automation and Insights in the Hospitality Sector

- Research question 5: Can LLMs be effectively utilized for the evaluation of customer reviews, and how can they revolutionize the hospitality sector by automating review responses and extracting actionable insights?
- Research statement 5: As proven by this study, LLMs have the potential to significantly support the hospitality sector by automating review evaluation, enhancing customer interactions, and providing valuable insights for business improvement.

Artificial intelligence is rapidly advancing, and its applications across various sectors of the economy are becoming increasingly numerous. Particularly with the arrival of LLMs, such as GPT-3.5 and ChatGPT in early 2023 [60], within a year, TV screens were filled with advertisements for AI-powered air conditioners, smartphones, personal assistants, and chatbots—even for communication with public services. While some advertisements may exploit the term “AI” for marketing purposes, many of these innovations genuinely have the potential to make everyday tasks more accessible to the public.

On the business side, companies are keen to create unique products that attract more customers or improve customer satisfaction. In highly competitive sectors such as tourism, the need for product differentiation is even greater [61]. This is where AI, and specifically LLMs, come into play. Ideas such as customer service chatbots, AI-driven booking systems, and models that detect spam in customer reviews [62], among others, all leverage AI, deep learning, and LLMs.

However, measuring customer satisfaction presents a unique challenge. Businesses can gauge satisfaction by reading reviews on social platforms, Google My Business, TripAdvisor, etc. For a tourism entrepreneur, reading and responding to reviews for an entire summer season could take as much time as another summer in front of a screen. The solution proposed in this research is the use and training of LLMs for sentiment analysis and classification of hotel reviews. In other words, an automated system would allow the entrepreneur to quickly identify critical reviews and respond to each. To achieve this, we used OpenAI's GPT omni family models and directly compared them with older methods, such as traditional NLP. Each model in the study was tasked, both pre- and post-fine-tuning, to perform sentiment analysis and classify reviews, predicting the customer rating. The results were encouraging, with the GPT-4o model leading in performance, and the GPT-4o mini being recognized for its cost-effectiveness.

Why, then, is it important for LLMs to understand customer sentiment? While LLMs are not initially designed for classification tasks, our results, along with numerous studies presented in the literature review, show that they are more than capable. The innovation of LLMs lies in generating text in response to a prompt. In our research, knowing the sentiment of a hotel guest, and having trained the appropriate LLM, we can use the same fine-tuned model to respond automatically to reviews on social media, TripAdvisor, and similar platforms. A fine-tuned LLM not only benefits from its general knowledge acquired during pre-training but also becomes specialized in a specific task, enabling it to automate many processes in the hotel and tourism sectors more broadly.

5.6. Limitations and Future Directions

In this study, we focused specifically on hotel reviews to evaluate the performance of advanced AI models in sentiment and emotion analysis within the hospitality sector. While this approach provided meaningful insights and practical recommendations for hotel decision-makers, the scope of our analysis is limited to a single domain within the broader tourism industry. Future research could explore the applicability of the methodologies presented here to other types of tourism-related content, such as restaurant reviews, travel blogs, or tourist attraction feedback. Such an expansion would enable a more comprehensive evaluation of AI models such as GPT-4 omni and BERT in addressing diverse challenges within the tourism ecosystem.

Additionally, our findings suggest that the fine-tuning and zero-shot learning approaches employed in this study have the potential for broader applications beyond the hospitality industry. These methodologies could be applied effectively to other domains of user-generated content, offering valuable tools for data-driven decision-making across various sectors. This cross-domain adaptability underscores the versatility of these approaches and highlights opportunities for further research and development.

Ethical considerations, particularly regarding confidentiality in fine-tuning AI models, were not directly addressed in this study but remain critical in the broader discourse on AI adoption. Fine-tuning often involves the use of proprietary or sensitive datasets, which raises concerns about data privacy, ownership, and compliance with regulations such as GDPR or CCPA. In the tourism industry, user-generated content, such as hotel reviews, may contain personally identifiable information (PII), emphasizing the need for robust anonymization techniques and ethical safeguards. While our research primarily focuses on model performance and practical applicability, future work should incorporate ethical frameworks to ensure the responsible development and deployment of AI technologies, fostering trust and transparency among users and stakeholders.

6. Conclusions

In conclusion, this study aimed to evaluate the effectiveness of advanced AI models, particularly LLMs such as GPT-4 omni and BERT, in sentiment analysis of hotel reviews, to compare their performance before and after fine-tuning, and to assess their cost-effectiveness in real-world applications. Our findings show that LLMs, especially the fine-tuned GPT-4 Omni, outperform traditional models such as BERT in sentiment classification and star rating prediction. While GPT-4 omni mini showed greater improvement after fine-tuning, the GPT-4 omni model achieved superior overall performance, highlighting the importance of model size in determining effectiveness.

Furthermore, we examined the financial and operational implications of fine-tuning LLMs. Our results show that smaller models, such as the GPT-4 omni mini, offer a cost-effective alternative without sacrificing performance, making them a viable choice for resource-constrained applications. This insight is critical for stakeholders in the tourism industry looking to balance performance with cost-efficiency.

In summary, this study reinforces the transformative potential of LLMs in the hospitality industry, not only for automating sentiment analysis but also for generating actionable insights from customer feedback. By adopting a balanced approach that takes both performance and cost into account, businesses in the tourism sector can leverage AI-driven tools to enhance customer experience, improve operational efficiency, and drive strategic decision-making.

Author Contributions: Conceptualization, K.I.R. and N.D.T.; methodology, K.I.R., N.D.T., and D.K.N.; software, K.I.R.; validation, K.I.R. and N.D.T.; formal analysis, K.I.R., N.D.T., and D.K.N.; investigation, K.I.R., N.D.T. and D.K.N.; resources, K.I.R.; data curation, K.I.R.; writing—original draft preparation, K.I.R. and N.D.T.; writing—review and editing, K.I.R., N.D.T., and D.K.N.; visualization, K.I.R.; supervision, D.K.N. and N.D.T. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data supporting the reported results can be found at ref [56].

Conflicts of Interest: The authors declare no conflicts of interest.

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